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ALY 6050

Module 5 Project

Using Linear Programming Models to Maximize Profits

Introduction

This report concerns the performance of four products in the first month of a company that is opening a new distribution center. With a monthly purchasing budget of $170,000 and new location allocated to only a certain amount of space for inventory, the aim is to maximize profits of these products when factoring in costs associated with the items. The four products being sold are pressure washers, go-karts, generators, and water pumps. The inventory space has 82 shelves that are 30 feet long and 5 feet wide, and the items are placed on pallets in the storage space.

Analysis

Mathematical Formulation of Problem

Decision Variables:

X1: The number of pressure washers to purchase.

X2: The number of go-karts to purchase.

X3: The number of generators to purchase.

X4: The number of water pumps to purchase.

Objective Function: To maximize profit of the items.

Optimal Solution: Z=(Price-Cost)x1 + (Price-Cost)x2 + (Price-Cost)x3 + (Price-Cost)x4

Z=($499.99 - $330)x1 + ($729.99 - $370)x2 + ($700.99 - $410)x3 + ($269.99 - $635/5)x4

Constraints:

- The monthly purchasing budget can't exceed the allocated $170,000.
- The inventory for the pressure washers and go-karts must be at least 30 percent of the total inventory.
- The company must sell at least twice as many generators over water pumps.
- The inventory space can't exceed 82 shelves with each shelf measuring out to 30 feet long and five feet wide. The pallets are 5X5 for the pressure washers and generators, 8X5 for the go-karts, and 5X5 for the water pumps that are stored in four cases.

Constraint Functions:

X1 + X2 + X3 + X4 <= 170,000

X1 + X2 >= 0.30 (X1 + X2 + X3+ X4)

X4 >= (2)X3

25X1 + 40X2 + 25X3 + 25/4X4 <= 82 * 30 * 5

2. Set up the linear programming formulation in an Excel workbook or R.

3. Use the Excel Solver or R to solve the problem and generate a sensitivity report.

A	B	C	D	E	F	G
Variables			Costs	Price	Profit	
x1	0		330	499.99	169.99	
x2	0.803322		370	729.99	359.99	
x3	394.9118		410	700.99	290.99	
x4	383.2117		127	269.99	142.99	
Objective						
Maximize	170000					
Constraints						
		Inequality	RHS			
1	170000	<=	170000			
2	289.188	>=	289.188			
3	54795.44	>=	789.8235			
4	12300	<=	12300			

Variable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
B2	x1	0	-1.42109E-14	169.99	1.42109E-14	1E+30
B3	x2	0.803322315	0	359.99	1.99835E-14	3.00945E-14
B4	x3	394.9117658	0	290.99	280.97	0
B5	x4	383.2116741	0	142.99	0	70.2425

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
B13		170000	1	170000	110958.3588	26437.30799
B14		289.188	-5.55112E-17	289.188	71952.06233	289.188
B15		54795.43728	0	0	54005.61374	1E+30
B16		12300	0	12300	2263.708027	4849.917775

4. Describe the optimal solutions obtained in the Word document. These will consist of the inventory level for all four products and the optimal monthly profit.

The optimal solution represents a particular set of values assigned to decision variables that either maximize or minimize the objective function, effectively optimizing it. In real-world optimization contexts, like those encountered in practical scenarios, there are usually numerous decision variables and a variety of constraints that these variables must satisfy (Evans, 2012). The decision variables used are the items that are being sold and the values associated with each item that will help achieve the objective function is maximizing profit of those items. Those values are calculated by subtracting the selling price of the items by the costs attached to the items that the company will pay for.

After initiating the Solver option in Excel, the calculations above show the optimal values of each item with the monthly purchasing budget in consideration. The generators and water pumps show a high optimal value of $394.91 and $383.21, respectively, while the go-karts barely register a value. The pressure washers have an optimal value of zero under the monthly purchasing budget and the constraints that were factored in.

The allowable increase refers to the maximum allowed increase to the decision variable's coefficient before causing the overall optimal solution to be revised. The generators were the only item that had a significant value that could not impact the overall optimal solution. The allowable decrease denotes the maximum allowable reduction in a variable's coefficient without affecting the current optimal solution. From the report, the water pumps were the only item that had a value large enough that the current optimal solution wouldn't be affected. When either the allowable increase or allowable decrease values for the changing cells are zero, it indicates the possibility of alternative optimal solutions. However, identifying these alternatives is not straightforward with Excel Solver. A problem is considered unbounded if the objective can be infinitely increased or decreased while maintaining feasibility (Evans, 2012).

5. One of the decision variables has an optimal value of zero. Use the Solver sensitivity report to determine the smallest selling price for that item so that this optimal zero solution value changes to a non-zero value.

The decision variable that has a monthly purchasing optimal value of zero is the pressure washers. Using the Solver sensitivity report the price of the pressure washers would need to be increased to $540.86 per item for the value to become a non-zero value. The issue with the price increase is that the go-karts, which also contain a very low optimal value, will then drop to zero. The values connection between those two decision variables is verified by the shadow price in the sensitivity report. The shadow price collectively influences how the objective function reacts to changes in constraints. If a constraint has positive slack which suggests surplus resources, the shadow price typically remains zero (Evans, 2012). The only constraint with a shadow value above zero is the connection between the pressure washer and the go-kart. This suggests that the optimal value in the pressure washer becoming a non-zero value will impact the optimal value of the go-kart.

6. In the word document explain whether, in addition to the $170,000 allocated to the purchasing budget during the first month, the company should allocate additional money.

If yes, how much additional investment do you recommend, and how much should the company expect its net monthly profit to increase?

Looking at the results from the Excel Solver it is evident that some items have a lot more value than others when looking at all factors like warehouse space, transportation costs, and promotional fees. Despite the go-kart's accruing the highest profit margins of all the items sold, it barely has any optimal value overall. One possible reason for this is since they take up more inventory space than the other items, they can't be sold in quantities like the other items can. Another possible factor is the promotional campaign that results in the pressure washers and go-karts together taking up at least 30 percent of the inventory space.

My recommendation would be not to allocate additional money for the monthly purchasing budget and to instead focus on the demand for its items overall. I would revise the inventory space percentages so there isn't an exact number of items like pressure washers and go-karts and so profits are not impacted by this. Instead of increasing the budget I would prefer to see how the adjustment of inventory space would affect the optimal values of each decision variable and look at whether this would generate no-zero values in each item. If the data shows that go-karts are not selling at the same numbers as the other items, then that would free up space for the other products that are selling more often. For example, if the water pumps are being purchased at higher rates, then this would increase space because they come in cases of four cases per pallet.

7. In the word document, explain whether you recommend that the company should rent a smaller or a larger warehouse. In any case, indicate the ideal size of your recommended warehouse in square feet, and indicate how much this change in the size of the warehouse will contribute to the monthly profit.

Based on the large pallet sizes of the go-karts and the lack of optimal values from half of the items, I believe that storage space is a concern for the company. If promotional fees that center on pressure washers and go-karts continue then a larger warehouse of 20,000 square feet (about four times the area of a basketball court) would be ideal for better profit margins among all the items. This would allow for 30 percent of the inventory to be allocated to the pressure washers and go-karts but can be edited if outcomes change and then more space will be beneficial toward potentially raising profits. The policy of selling twice as many generators as water pumps can also remain in place and can also be revised if outcomes change with more warehouse storage space.

One concern with a larger warehouse is how this could affect the costs of the products being sold. Would moving to a new warehouse increase the transportation costs attached to each item? Are there additional costs that will change or rise from having more inventory space in the future? Will having more inventory space be a burden on the monthly budget where extra funds and resources will need to be used toward areas like maintenance of the building or other added labor costs? These questions should be thoroughly researched and answered so sales don't decrease overall and therefore affect the company's profit margins.

Conclusion

There are some areas of focus that show positive correlations between the decision variables and constraints that will continue to generate profit for the company. There are also aspects of the company where there is some room for improvement so that all items being sold have immense value. I recommend continuing to monitor the current business model and make necessary revisions before deciding to take the next crucial step in whether the budget needs to be increased or that more inventory space is required for future financial gains.

References

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